The Impact of Insomnia on Visuospatial Attention and Reaction Times

Introduction:

Visuospatial attention is essential for distinguishing relevant visual stimuli from distractions. Although sleep is essential for cognitive function, it is unknown how it affects visuospatial attention. This study investigates the relationship between reaction times (RTs) and sleep disorders, specifically insomnia. Using a behavioral experiment, participants completed a visual detection task under different distractor conditions, allowing for an assessment of attention and processing speed. Additionally, the association between personal factors including age, IQ, and sleep duration and attentional performance was investigated.

The study examines the following: (1) how distractor levels affect RTs and accuracy; (2) how sleep duration and distractor effects are related; and (3) how normal and insomniac people differ from one another. Group type was the between-subject factor and distractor level was the within-subject factor in this mixed factorial design.

Method:

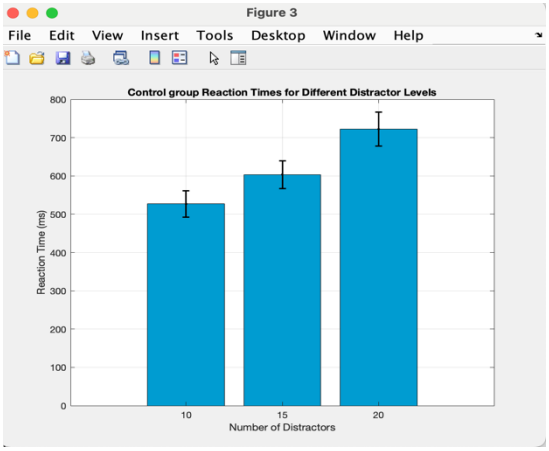
In this study, 35 controls and 28 insomniacs performed a visual perception task with 10, 15 and 20 distractors. Reaction times (RTs) and accuracy were recorded, and only correct responses were included in the RT analysis. Age, IQ and sleep duration were also collected.

Within each group, repeated measures ANOVAs were conducted to assess the distractor effect and pairwise comparisons with Bonferroni corrections were applied. A mixed ANOVA compared RTs between groups taking into account distracter levels and group type.

Pearson correlation explored the influence of age, IQ and sleep duration on the distractor effect (RT difference between 20 and 10 distractors), while multiple regression analysis assessed their combined contribution.

Results:

The analysis assessed the effect of the number of distractors on response times and accuracy in the **control group**. There was a significant effect of distractor level on reaction times, **$F(2, 102) = 6.5$, $p = 0.0022$,** with times increasing as distractors increased **(526.78 ms, 603.44 ms and 722.23 ms for 10, 15 and 20 distractors).** However, the effect on accuracy was not significant, **$F(2, 102) = 0.5$, $p = 0.6071$,** with accuracy remaining constant across conditions **(93.21%, 94.14% and 93.42%).** Post-hoc comparisons confirmed significant reaction time differences between all distractor levels, but accuracy did not change.



The effect of distractors on reaction times and accuracy was also analyzed in the **insomnia group**. A significant effect on reaction times was found, with times increasing as distractors increased, **$F(2, 81) = 10.29$, $p = 0.0001$ (518.16 ms, 639.27 ms and 788.74 ms for 10, 15 and 20 distractors).** However, accuracy remained constant across conditions, **$F(2, 81) = 0.15$, $p = 0.8584$ (94.91%, 95.00% and 95.35%).** Post-hoc comparisons confirmed significant reaction time differences, but accuracy did not change.

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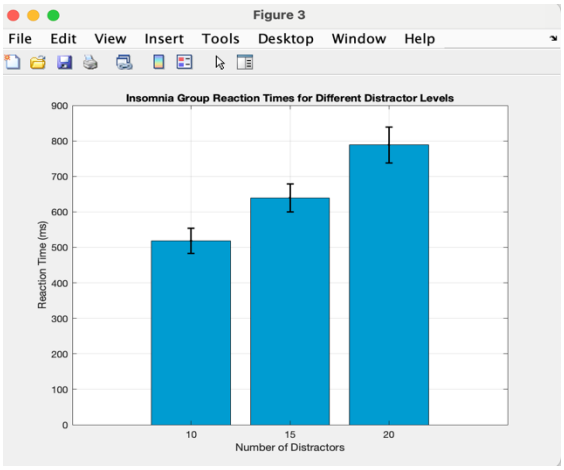
A factorial mixed ANOVA was performed to examine the effects of group (insomnia vs control) and distractor levels (10, 15 and 20) on reaction times.

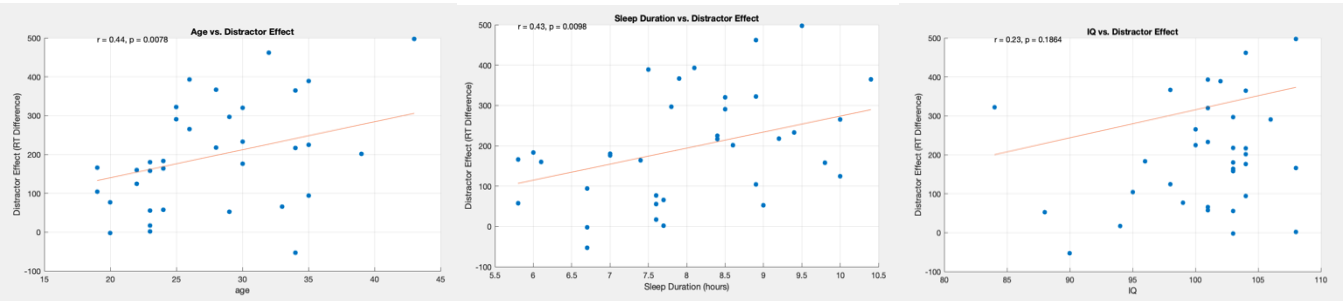


The results showed that there was no significant difference between groups **(F(1,183) = 0.89, p = 0.347)** but there was a significant effect of distractor levels **(F(2,183) = 16.61, p < 0.001),** with reaction times increasing as distractors increased. No significant interaction was found **(F(2,183) = 0.43, p = 0.6491).**

These findings suggest that while distractor levels affect reaction times, both groups follow a similar pattern and no additional cognitive impairment was observed in the insomnia group.

Correlation analysis showed that age **(r = 0.4424, p = 0.0078)** and sleep quality (**r = 0.4305, p = 0.0098)** were significantly associated with reaction time, suggesting that older age and better sleep quality influence reaction speed. However, IQ **(r = 0.2287, p = 0.1864)** did not show a significant effect.

Regression analysis confirmed these results. Age significantly predicted reaction time, **β = 7.21, p = 0.0505,** indicating that reaction time increases with age. Similarly, sleep quality significantly predicted reaction time, **β = 39.72, p = 0.0249,** suggesting that better sleep is associated with faster reactions. In contrast, IQ did not significantly predict reaction time (**β = 5.49, p = 0.1606).**



Implications:

Although reaction times rose with additional distractions, both groups appeared to follow a similar pattern, as indicated by the lack of a significant interaction between group and distractor level. This implies that insomnia may still have an impact on more general cognitive functions even if it does not directly heighten the distractor effect.

The importance of sleep in preserving cognitive function is highlighted by the strong correlation found between reaction time and sleep quality. Future studies should look into whether sleep interventions enhance cognitive performance because faster reactions are linked to higher sleep.

The brain mechanisms behind attentional differences in people with sleep disorders may be better understood with the use of neuroimaging techniques like fMRI or EEG. This can result in focused treatments to lessen the cognitive deficits brought on by sleeplessness.

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